Phoenix at AraGenre 2026: Transductive Bias Correction and Multi-View Label Representations for Zero-Shot Arabic Genre Classification

Nguyen Minh Chi
Independent Researcher
minhchi1804@gmail.com

Abstract

AraGenre 2026 asks systems to classify Arabic texts into a hierarchical genre taxonomy whose evaluation-specific genres are unseen. The supplied genre menu provides names, taxonomy structure, and English definitions, but no labelled examples of those genres. We describe a two-stage, transductive zero-shot system. A multilingual encoder scores each text against genre representations built by mean-pooling several template-derived views of each definition, and a transductive per-genre z-normalization reduces definition-specific bias before a candidate shortlist is formed. Qwen3-8B reranks the shortlist by calibrated candidate log-probabilities – a content-free baseline mitigates the LLM's strong position prior – and the two score sets are fused; family-restricted sibling reranking, a macro-F1 plug-in decoder, and a margin-gated 9-wide pool add the final gains. On the official 27,972-instance test set the system reaches **0.5800** Hierarchical Macro F1 (broad 0.749, specific 0.411). In the main forward build-up, per-genre normalization gives the largest increment ($+0.095$ Hierarchical Macro F1); a separate encoder-only diagnostic attributes $+0.082$ to LLM reranking. Calibrated logprob fusion ($+0.020$), multi-view labels ($+0.024$), and the decoding refinements ($+0.020$) provide the remaining gains.

1 Introduction

AraGenre is a shared task on Arabic genre classification at fine granularity (El-Haj et al., 2026). A system must predict a broad genre (six classes) and a specific genre (74 classes on the test split) for each Arabic document, where every genre is accompanied by a short *English* definition. The specific genres of the test set are disjoint from those seen in the labelled data, so a system must generalise from the supplied names, hierarchy, and definitions without labelled examples of the target genres. This definition-guided, zero-shot setting, combined with a macro-averaged metric over a large and heavily imbalanced label space, makes the task unusually sensitive to *label-prior bias*: the tendency of both encoders and LLMs to prefer certain labels or positions regardless of the input.

Our system is a two-stage pipeline. Stage 1 is multilingual dense retrieval (Wang et al., 2024a): each text is scored against genre representations built by mean-pooling several template-derived *views* of each definition (canonical, source type, topic, communicative function), and the similarity matrix is calibrated with a transductive per-genre z-normalization. Stage 2 is LLM reranking: Qwen3-8B (Yang et al., 2025) scores the calibrated candidates by the log-probability of single-digit answer tokens, corrected by a content-free contextual baseline (Zhao et al., 2021), and fused with the retrieval scores. Two decoding refinements and a margin-gated 9-wide pool complete the pipeline. The broad genre is never predicted directly; it is read off the chosen specific genre.

Our official test-set score is **0.5800** Hierarchical Macro F1 (broad 0.749, specific 0.411). Two intuitive alternatives – broad-first hierarchical decoding (0.2262) and uniform-marginal calibration (0.4974) – underperform. No instance-level test labels were available to any component; calibration and shortlisting are transductive over the *unlabelled* test inputs, while aggregate leaderboard feedback was used to select some settings. Code is available at https://github.com/geebabe/ARAGENRE.

2 Background

Each instance is a record $\{\mathrm{id}, \mathrm{text}\}$ of Arabic web text; a shared genre menu supplies, for every allowed genre, a record of broad genre, broad definition, specific genre, and specific definition (English paragraphs). The input is Arabic while all definitional evidence is English, so the task is intrinsi-

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 261–265
October 28-29, 2026 ©2026 Association for Computational Linguistics

	train	dev
instances	140	110
broad genres	5 (no Religious)	6
specific genres	7	6

Table 1: Public development splits. The hidden test split contains 27,972 instances, 6 broad and 74 specific genres; specific genres are disjoint across all splits.

cally cross-lingual and definition-guided. Table 1 summarises the official splits: dev is controlled and synthetic, test is naturally occurring and noisy, specific genres are disjoint across splits, and training contains no Religious instances. A striking test-set property is its length: a median of 46 words, with 35.6% of items under 150 characters, which likely contributes to specific-level errors.

Our approach builds on three lines of work. Definition-guided zero-shot classification scores label descriptions directly (Basile et al., 2022), and aggregating several contextualised label formulations stabilises scoring (Kumar et al., 2024); at high cardinality, retrieval-augmented LLM ranking over a compact candidate set is effective (Pattnaik et al., 2025). LLM predictions over large label sets suffer position and format biases, mitigated by *contextual calibration* (Zhao et al., 2021) and its domain-level extension (Fei et al., 2023); decoding-free selection scores candidate tokens by log-probability (Ma et al., 2025), since first-token probabilities of generated answers are unreliable for instruction-tuned models (Wang et al., 2024b). Because the official metric is macro-F1, we decode with a metric-aware plug-in rule rather than the per-item argmax (Waegeman et al., 2014; Dembczynski et al., 2011; Parambath et al., 2014). The song-lyric genres also touch Arabic dialect identification, where labels are often non-exclusive (Abdelali et al., 2021; Keleg and Magdy, 2023).

3 System

All stages operate without gold labels; the inputs are the Arabic texts and the supplied genre menu, including genre names, taxonomy structure, and definitions: *multi-view retrieval* → *per-genre z-norm* (transductive) → *calibrated logprob reranking* (5- and 9-wide) → *sibling rerank, macro-F1 decode, margin-gated 9-wide* → *broad read off the chosen specific*.

3.1 Multi-view label representations

Of the 74 test genres, 45 (60.8%) share a *source type* and differ only by *topic*: 20 job ads, 13 book descriptions, 8 student-writing genres and 4 textbook levels. In the supplied definition the discriminating word is buried in shared boilerplate ("Job advertisement texts related to *X*, typically describing responsibilities…"), so siblings such as `sales_jobs` and `marketing_jobs` collapse together. We therefore encode each genre under up to four *template-derived views* and mean-pool them: the *canonical* definition verbatim; a *source* view ("This text is a job advertisement"); a *topic* view ("The subject matter is sales"), free of shared boilerplate; and a *function* view ("The author is advertising an open position and listing its duties"). Views are derived mechanically from the taxonomy's naming structure (suffixes such as `_jobs`), not generated by a model. A genre's representation is the renormalised mean of its views,

$$\bar{v}_g = \frac{\sum_{v \in \mathcal{V}_g} e(v)}{\left\| \sum_{v \in \mathcal{V}_g} e(v) \right\|}, \tag{1}$$

where $\mathcal{V}_g$ is the view set of genre g and $e(\cdot)$ is the unit-norm encoder embedding.

3.2 Calibrated retrieval

Queries are encoded with the E5 instruct convention ("`Instruct:` *task* `Query:` *text*") and scored by cosine similarity against the pooled genre representations, giving an $N \times 74$ matrix S. Uncalibrated, this matrix is dominated by a few "attractor" definitions: `msa_poetry` receives 25.0% of all predictions and the Creative broad class 48.8%, starving the other labels under macro averaging. We reduce this per-genre bias with a transductive per-genre z-normalization over the *unlabelled* test inputs,

$$\begin{aligned} z_{ij} &= \frac{s_{ij} - \mu_j}{\sigma_j + \epsilon}, \\ \mu_j &= \frac{1}{N} \sum_i s_{ij}, \\ \sigma_j^2 &= \frac{1}{N} \sum_i (s_{ij} - \mu_j)^2 . \end{aligned} \tag{2}$$

After calibration all 74 genres are used and the maximum class share falls from 25.0% to 4.0%; candidate shortlists of width 5 and 9 are taken from the calibrated scores.

3.3 Calibrated logprob reranking

The reranker, Qwen3-8B with thinking disabled, lists the shortlisted candidates with their definitions and is *prefilled* with "`Answer:` ". We explicitly score only the candidate digits at the next-token position. Since $k \leq 9$ each candidate is a single token, full-sequence likelihood and first-token log-probability coincide (Ma et al., 2025; Wang et al., 2024b); the score is a log-softmax over the candidate digits only. The LLM carries a strong position prior – with the text replaced by a placeholder, it still prefers option 1 on 51.4% of items (vs. $1/k$=20%) – so we subtract a content-free contextual baseline (Zhao et al., 2021):

$$\ell_j(x) = \log \frac{\exp(\text{logit}_j)}{\sum_{c \in C_x} \exp(\text{logit}_c)}, \qquad L_j(x) = \ell_j(x) - \frac{1}{3} \sum_{\tau \in \mathcal{T}} \ell_j(\tau), \tag{3}$$

where C_x is the candidate set and $\mathcal{T} = \{\text{N/A}, \varnothing, \texttt{[MASK]}\}$, each placeholder scored over the same prefilled shortlist ($C_\tau = C_x$). The corrected LLM scores and the calibrated retrieval scores are each row-standardised within a shortlist and fused:

$$S_j(x) \;=\; \alpha\, \tilde{E}_j(x) \;+\; (1-\alpha)\, \tilde{L}_j(x), \tag{4}$$

Fusions were generated offline from cached log-probabilities, but the final α=0.25 was selected using official Codabench feedback: 0.5316 at α=0 versus 0.5353 at α=0.25. For metric-aware decoding, a temperature-T softmax turns the fused shortlist scores into a per-item pseudo-posterior,

$$p_j(x) \;=\; \frac{\exp(S_j(x)/T)}{\sum_{c \in C_x} \exp(S_c(x)/T)}, \qquad T = 1. \tag{5}$$

The fused argmax is unchanged by this transformation. The broad genre is the unique broad of the selected specific genre and is *never* predicted directly.

Refinements. Three components complete the system. (i) *Sibling reranking*: when an item's top-2 fused candidates are siblings in one source-type family (Section 3.1), a narrow second prompt listing only that family's shortlisted members (re-indexed $1..k$, $k \leq 5$) – shared source type stated as given, discriminating axis named – is scored with the same content-free calibration; a single presentation order suffices – averaging forward and reversed orders was measured to lose on every offline gate. The corrected sibling scores are row-standardised and blended with the main scores for sibling candidates, $S'_j = (1-\beta)S_j + \beta\tilde{L}^{\text{sib}}_j$, with $\beta = 0.5$; non-sibling scores are unchanged. (ii) *Macro-F1 plug-in decoding*: we recompute Eq. (5) from S' and apply coordinate ascent with $\lambda = 1.0$ to optimize the plug-in macro-F1 estimate over the unlabelled test inputs, reallocating assignments toward starved classes without imposing a target marginal (unlike Sinkhorn). (iii) *Margin-gated 9-wide widening*: candidates are also scored over a 9-wide pool (single-digit tokens cap $k \leq 9$); the 9-wide winner is adopted only when it lies outside the 5-wide pool (encoder rank 6–9) and wins by margin $> \tau$=0.5, keeping confident 5-wide decisions.

4 Experimental Setup

No component is trained on the supplied labels; transductive quantities (Eq. (2)) are computed over the unlabelled test inputs only, and no additional task-specific corpora were used. Development-based selection was rejected after an uncalibrated retrieval+Qwen3 system reached 0.9488 Hierarchical Macro F1 on the controlled dev set but only 0.4200 on the noisy test set. Instead, the final fusion weight α=0.25 and widening threshold τ=0.5 were selected using aggregate scores from official Codabench submissions; τ was chosen from the submitted ladder $\{1.2, 0.5, 0.0\}$. This selection used only aggregate scores returned through the official shared-task evaluation process; no instance-level test labels or other test-set annotations were accessed. Preprocessing is none beyond sentence-transformers tokenization (inputs truncated at the encoder's 512-token maximum); LLM items are length-sorted and processed in chunks of 128. Models: `intfloat/multilingual-e5-large-instruct` (Wang et al., 2024a) and `Qwen/Qwen3-8B` (Yang et al., 2025) via `sentence-transformers` and `transformers` on Modal serverless NVIDIA L4 GPUs. Hyperparameters: candidate pools k=5 and k=9, fusion α=0.25, margin gate τ=0.5, content-free baseline $\mathcal{T}$ averaged over three placeholders. The official metric is Hierarchical Macro F1 = mean(broad macro F1, specific macro F1); all scores below are official Codabench submissions from the final evaluation phase.

5 Results

Table 2 reports a forward build-up of the final system. The main block starts from uncalibrated Qwen3 greedy selection and incrementally adds each component. The final row is a separate encoder-only diagnostic used to measure the LLM's contribution. Every row is a real hidden-test submission.

System	broad	specific	hier.
Uncalibrated Qwen3 greedy	—	—	0.4200
+ per-genre z-norm	0.664	0.366	0.5150
+ calibrated logprob fusion	0.688	0.383	0.5353
+ multi-view labels	0.727	0.392	0.5597
+ sibling rerank + macro-F1 decode	—	—	0.5670
+ margin-gated 9-wide widening	0.749	0.411	**0.5800**
Calibrated encoder only (diagnostic)	0.566	0.299	0.4326

Table 2: Official test-set results (27,972 instances). The main block is a forward build-up; the encoder-only row is a separate diagnostic comparison. Deltas are order-dependent; broad and specific scores are rounded to three decimals, while hierarchical scores use the unrounded values. — denotes unrecorded broad/specific scores.

Calibration gives the largest build-up gain. Holding Qwen3 greedy selection fixed, per-genre z-normalization raises Hierarchical Macro F1 by $+0.095$ (0.4200→0.5150). Broad and specific scores for the uncalibrated submission were not recorded, so no component-level specific gain is claimed. Calibration adds no information; it reduces the attractor bias that concentrates predictions on a handful of definitions, expanding coverage from a few dominant labels to all 74 genres.

The LLM is complementary, not redundant. Adding the reranker over the calibrated shortlist gains $+0.082$ hier. and $+0.067$ specific macro F1 in the separate encoder-only comparison (0.4326→0.5150 and 0.299→0.366). Calibrated logprob fusion then gains a further $+0.020$ (0.5150→0.5353). Multi-view label representations add $+0.024$ (0.5353→0.5597), lifting broad macro F1 from 0.688 to 0.727. Decoding refinements add the final two increments: sibling reranking with the macro-F1 plug-in decoder ($+0.007$, 0.5597→0.5670) and the margin-gated 9-wide widening ($+0.013$, 0.5670→0.5800; broad 0.749, specific 0.411).

Two intuitive designs failed. Top-down decoding (broad before specific) scored 0.2262: broad definitions are too abstract to embed against (0.385 broad accuracy), and a wrong broad makes the specific unrecoverable. Sinkhorn calibration toward a uniform 74-way marginal moved the broad mix almost exactly onto the uniform reference and lost 0.018 (0.4974): the test set is *not* uniformly distributed, so we reject any method rewarding a flatter output distribution.

5.1 Error analysis

The residual gap is broad-vs-specific: broad macro F1 0.749 versus specific macro F1 0.411. Specific macro F1 also trails weighted F1 (0.438) and accuracy (0.441), indicating weaker performance on less frequent classes.

Short texts provide limited evidence. With a 46-word median and 35.6% of items under 150 characters, many documents contain little topical evidence for a 74-way distinction.

Sibling overlap is visible on development data. By share of the development set, all six errors map *Advice Columns* to *Motivational Writing*, illustrating ambiguity among related genres without establishing the test-set confusion distribution.

The candidate tail is still active. The calibrated pick-by-rank distribution is nearly flat at the 5-wide cutoff (ranks 4–5 take 20.4% of picks): 12.6% of items prefer a candidate at encoder rank 6–9. The margin-gated widening adopts a subset ($+0.013$ hier., 9.5% of labels changed at τ=0.5), so the tail is likely not yet exhausted.

6 Conclusion

Our transductive zero-shot system reaches 0.5800 Hierarchical Macro F1 (broad 0.749, specific 0.411), with complementary gains from per-genre normalization, calibrated logprob fusion, multi-view labels, and metric-aware decoding.

Limitations

Short inputs and five dialect-only lyric genres remain challenging. Because refinement choices, α, and τ used aggregate official-submission feedback, the configuration is leaderboard-selected; the reported hidden-test score and component gains should therefore be interpreted as partly conditioned on test-set model selection rather than as

fully independent estimates of generalisation. Future work will explore domain-context calibration (Fei et al., 2023), retrieval ensembles, grade-level views, and multi-token candidate pools.

Ethical Considerations

Our system assigns genre labels to Arabic web texts using organiser-provided shared-task inputs and publicly released pretrained models. Cross-lingual encoders and LLMs can inherit social, linguistic, and regional biases, which may contribute to uneven performance across dialects and less frequent genres. Genre predictions should not be interpreted as judgments about a text's author and should not be used for high-stakes moderation or profiling without further validation. We did not attempt to identify individuals and do not release dataset-derived text.

Acknowledgments

We thank the AraGenre organisers and open-model developers.

References

Ahmed Abdelali, Hamdy Mubarak, Younes Samih, Sabit Hassan, and Kareem Darwish. 2021. QADI: Arabic dialect identification in the wild. In *Proceedings of the Sixth Arabic Natural Language Processing Workshop*, pages 1–10, Kyiv, Ukraine (Virtual). Association for Computational Linguistics.

Angelo Basile, Marc Franco-Salvador, and Paolo Rosso. 2022. Unsupervised ranking and aggregation of label descriptions for zero-shot classifiers. In *Natural Language Processing and Information Systems*, volume 13286 of *Lecture Notes in Computer Science*, pages 119–126, Cham. Springer International Publishing.

Krzysztof Dembczynski, Willem Waegeman, Weiwei Cheng, and Eyke Hüllermeier. 2011. An exact algorithm for F-measure maximization. In *Advances in Neural Information Processing Systems*, volume 24. Curran Associates, Inc.

Mo El-Haj, Saad Ezzini, Shadi Abudalfa, Mustafa Jarrar, Nguyen Minh Chi, and Nguyen Minh Quan. 2026. AraGenre 2026: A hierarchical definition-guided Arabic genre classification shared task. In *Proceedings of the 4th Arabic Natural Language Processing Conference (ArabicNLP 2026)*, Budapest, Hungary. Association for Computational Linguistics.

Yu Fei, Yifan Hou, Zeming Chen, and Antoine Bosselut. 2023. Mitigating label biases for in-context learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14014–14031, Toronto, Canada. Association for Computational Linguistics.

Amr Keleg and Walid Magdy. 2023. Arabic dialect identification under scrutiny: Limitations of single-label classification. In *Proceedings of ArabicNLP 2023*, pages 385–398, Singapore (Hybrid). Association for Computational Linguistics.

Sachin Kumar, Chan Young Park, and Yulia Tsvetkov. 2024. Gen-Z: Generative zero-shot text classification with contextualized label descriptions. In *The Twelfth International Conference on Learning Representations*.

Mingyu Derek Ma, Yanna Ding, Zijie Huang, Jianxi Gao, Yizhou Sun, and Wei Wang. 2025. Inferring from logits: Exploring best practices for decoding-free generative candidate selection. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 33125–33144, Vienna, Austria. Association for Computational Linguistics.

Shameem A. Puthiya Parambath, Nicolas Usunier, and Yves Grandvalet. 2014. Optimizing F-measures by cost-sensitive classification. In *Advances in Neural Information Processing Systems*, volume 27. Curran Associates, Inc.

Anup Pattnaik, Sasanka Vutla, Hamvir Dev, Jeevesh Nandan, and Cijo George. 2025. Scalable and cost effective high-cardinality classification with LLMs via multi-view label representations and retrieval augmentation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 1955–1969, Suzhou (China). Association for Computational Linguistics.

Willem Waegeman, Krzysztof Dembczyński, Arkadiusz Jachnik, Weiwei Cheng, and Eyke Hüllermeier. 2014. On the Bayes-optimality of F-measure maximizers. *Journal of Machine Learning Research*, 15(103):3513–3568.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024a. Multilingual E5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Xinpeng Wang, Bolei Ma, Chengzhi Hu, Leon Weber-Genzel, Paul Röttger, Frauke Kreuter, Dirk Hovy, and Barbara Plank. 2024b. "My Answer is C": First-token probabilities do not match text answers in instruction-tuned language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 7407–7416, Bangkok, Thailand. Association for Computational Linguistics.

An Yang, Anfeng Li, Baosong Yang, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. 2021. Calibrate before use: Improving few-shot performance of language models. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 12697–12706. PMLR.